Technology, Environment, and Society

Definitions

Technology

According to Webster's dictionary, technology is defined as a description of the arts.

General definition of technology
Technology is a technical means that involves the systematic application of organized knowledge, tools, and materials for the extension of human faculties.

Technology is the driving force of societal change.

Technological innovation
- Creative. feasible ideas
- Practical application
- Diffusion through society

Environment
The environment is defined as the immediate surroundings that support life and sustain various human activities. The surroundings comprise
- Biotic or living things: plants, animals, microorganisms
- Abiotic or non-living things: land, water, air, etc.

Society
Society is people living together in communities.

Chapter 2. Technology and society

2.1 Different types of technology

Labor-based technology

- Labor/equipment mix technology: a technology that gives priority to labor, supplemented with appropriate equipment
- Advantages
  - Creation of more employment (mostly unskilled)
  - Reduction of environmental impact
  - Increased use of associated local resources, contributing to the local economy.

Labor-intensive technology

- Involvement of large numbers of workers to produce goods or services
- High labor cost relative to capital
- Physical and manual job
- Labor-intensive industries include restaurants, hotels, agriculture, and mining
- Advantage: Control of expenses during market downturns by controlling the size of the employee base

Appropriate technology

- Technology that is appropriate to the environmental, cultural, and economic situation
- Requirement of fewer resources, as well as lower cost and less impact on the environment.

- Considered to be suitable for use in developing nations or underdeveloped rural areas of industrialized nations, which they feel cannot operate and maintain high technology
- Usually labor-intensive
- Some appropriate technologies
    - Information and communication technology
    - Construction
    - Solar cells, biogas, biofuel, wind power, micro hydro
    - Smokeless and wood conserving stoves
    - Rainwater harvesting, fog collection

2.2 Levels of technology

1. Low-level technology

- Up to 3200 BC
- Tools and machines developed by earlier human beings
- Primitive tools and simple machines that served as the foundations for other tools and machines
- Type of tools
    - Natural tools: no modification, naturally available, e.g. stone piece
    - Adapted tools: modified in size and shape, e.g., sharpened bone at one end
    - Manufactured tools: developed tools, e.g., arrow, bow, spear
- Human and animal-powered machines

2. Intermediate level technology

- From 3200 BC to the Industrial Revolution in some countries, to the present in most countries

- Tools that are more sophisticated or complex than those currently in use in a developing nation, but still much less costly, or more accessible, than those tools that would be used in a developed nation
- Improvement/modification of primitive tools and machines
  - Axe and hammer made of metal
  - Compound or complex machines powered by humans, animals, or other forces
  - Steam power, diesel engine, steam turbine
- Civilizations in this technological period

3. **High-level technology**

- Technology of the post-industrial era
- More sophisticated tools and machines
- Prevalent in most Western societies
- Sources of power: no human power, e.g., electric power, mechanical power, steam power
- Three subdivisions
  - Assembly line: a group of complex machines working in conjunction
  - Automation
  - Computer technology

2.3 Technology as a curse or as a blessing

- Handling of technology decides whether it is a curse or a blessing.
- Technology for betterment of life: blessing, technology for negative effects, e.g., destruction: curse

e.g., blessing: material things, airplane for easy transport, easy life, freedom from heat/cold, great human achievements
curse: airplane to carry bombs for war, pollution, and other destructive effects

2.4 Technology is now irreversible

- The development of technology: a result of the development of human civilization.
- From the early to today's modern technology, technological development has continued and obviously not reversed.
- Developing new technology, knowledge, and resources is needed. But to reverse technology, we can have the resources of the old days, but knowledge cannot be.

2.5 Technology creates the opportunity for society to change

Families and modern technology

- Due to technology, people are migrating for jobs, career, and better opportunities, breaking the tradition of the joint family
- A greater number of working women
- Sense of companionship is taking place in society among family members

Economy

- Change in the economy due to technology, broader market, wider area of the global economy
- High productivity due to technology

- High consumption due to low prices

Politics

- By using technology, more resources are used to generate wealth, and wealth gives power.
- Periodic reorganization of political forces due to technology
- Technology occasionally becomes a political asset

Education

- Technology development processes started because of education.
- Easier methods of learning, e.g., audio-visual aids
- Easy distribution of information distribution
- Technology has made education essential to earn a living.
- Models of education
    - Classical model: history, literature, philosophy, language, focus on past achievements
    - Religious model: focus on holy books of religion
    - Managerial model: focus on vocational education
    - Humanistic model: making people more human, focusing on arts and science

Religion

- Technology helps to unmask old social problems.
- For many people in modern societies, religion is neither good nor bad but simply irrelevant, given the many alternative ways to find meaning in various forms of cultural pursuits, ethical ideals, and lifestyles.

- Technology has changed spiritual beliefs into business opportunities.

2.6 Importance of technology in controlling prices

- Increase in the production rate at a low price with the help of modern tools like automation, management techniques, etc.
- More consumption of goods due to price reduction. According to the supply-demand chain, a low price is due to more consumption

  For example, electronic goods are cheaper now than yesterday.

Interaction between technology and the labor force

- With the development of technology, the use of muscle power is decreasing day by day
- The automatic system, robots, etc. displacing the labor force.
- Creation of new types of job opportunities requiring fewer qualified people in smaller numbers

2.7 Society's control over technology

- Technology should be adaptable to society according to social status and needs

  E.g., sustainable development for rural society, dynamic technologies for urban society
- If technology does not cope with the social requirement, then it goes out.

The benefits of society from new technological inventions

- Acceleration of social growth
  e.g., social transformation due to the invention of steam power, petrol cars, diesel engines, increase in interaction due to the invention of television, computers, airplanes

Technological innovation can unmask the old social problems

- Unmasking of cast discrimination and superstitions by education with the help of modern technology
- Unmasking chronic social problems by regular advertisement and awareness programs using modern technology like television, the internet, etc.

2.8 Impact of industrialization on societies that are not yet technologically developed

- **Positive:** opportunities, growth in trade and economy, infrastructure development, acceleration of human civilization
- **Negative:** water pollution, air pollution, noise pollution, ozone layer depletion, deforestation, etc.

Shifts in employment opportunities

- Creation of new kinds of jobs and businesses due to transportation, communications, and computers
- International competition in businesses

- Shifting from farming in rural areas to industrial jobs in cities and suburbs in Western countries.
- Change in production and employment patterns because of technological advances, increased levels of world trade, and a rapid increase in the demand for services.
- Change in the education pattern due to the shift in employment opportunities.

Chapter 3. Environmental issues

3.1 Introduction

Introduction to Ecology and Ecosystem

Ecology
- Oikos: home or surrounding, logos: study
- **Ecology:** The Science of the interrelationship between organisms and their relationship with environment

Ecosystem
- A natural unit that consists of biotic communities and their abiotic environment
- Basic functional unit in ecology, Types: Freshwater, grassland, marine, desert.

Characteristics of an ecosystem
1. Biotic component: producers (green plants), consumers (animals), decomposers (microorganisms)
2. Abiotic component: air, water, soil
3. Energy flow: The sun is the main source of energy
4. Matter
5. Interrelationship
6. Biological integration
7. Flexibility
8. Ecological regulation

Human impact on the environment/ecosystem
- Destruction or modification of habitat
- Overexploitation for commercial, scientific, and educational purposes
- Overgrazing for domestic animals
- Change in arable land
- Forestry
- Traditional rural practice
- Industrialization, Urbanization
- Mining and quarrying
- Pressure from introduced plants
- Population pressure
- Use of drugs and chemicals
- Destruction of ecological balance

Environmental sanitation
- Cleaning of the environment

Environmental sanitation includes the following:
1. Collection and disposal of refuse and sewage from houses, buildings, and other public places
2. Proper ventilation for the control of indoor air pollution: fresh air circulation
3. Sufficient light in the buildings for healthy conditions of the human body
4. Heating

Local environmental issues: water pollution, air pollution, noise pollution, solid waste pollution, deforestation, land degradation

Global environmental issues: Global warming, Acid rain deposition, Ozone layer depletion.

3.2 Water pollution

Water pollution: presence of various types of impurities that tend to degrade its quality and either constitute a health hazard or otherwise decrease the utility of water

Sources of water pollution

- Natural: Soil erosion, solutions of minerals in water, rainwater, storms, earthquakes, seawater intrusion, dust/dirt falling from the atmosphere, deposition of animal wastes and fallen leaves, etc.
- Man-made: Due to agriculture, sewage, waste, and industry

Sewage: Liquid waste from the community contains 99.9% water, 0.1% solids (organic/inorganic matter, disease-producing organisms)

Types of pollutants

- Pathogenic organisms
- Oxygen-demanding substances
- Plant nutrients: Nitrogen and Phosphorus
- Toxic organic chemicals: pesticides
- Poisonous inorganic chemicals
- Oil
- Thermal pollution (Heat): from power plants
- Sediment
- Radioactive substances
- Others: color, odor, taste

Impact of water pollution

- Health hazard due to the presence of pathogenic bacteria from domestic sewage, toxic materials, and industrial waste
  Water-borne diseases: typhoid, cholera, dysentery, infectious hepatitis
- Economic loss: disturbance of recreation, aesthetics, agriculture, industry, and property
- Impact on aquatic and plant life

Prevention of pollution

- Treatment of sewage
- Treatment of industrial waste
- Providing training and technical facilities in the industry to treat wastewater
- Not using a water source for discharging sewage
- Rules and regulations for controlling pollution
- Proper planning of towns

Sewage disposal method

1. Natural methods

- Dilution: discharging into a water course, e.g., sea, river, or lake, self-purification in due course of time
- Land treatment: spreading sewage on land, two ways: filtration, sewage farming

2. Artificial method: Sewage treatment method for the removal of suspended solids, pathogens

Cause, Effect, and remedial measures of various water pollutants

Physical

Impurity/pollutants	Cause	Effect	Remedial measures
1. Suspended solids (turbidity)	Clay, silt, organic matter, inorganic matter, minerals, algae, fungi	Turbidity, color, odor	Treatment: Settling, coagulation, filtration
2. Color	Dissolved organic matter, inorganic matter, minerals, and industrial waste	Objectionable from an aesthetic and psychological point of view, no health effect	Treatment method, such as aeration, treatment with activated carbon, oxidation of organic matter
3. Taste and odor	Dead or living microorganisms, dissolved gases, e.g., H_2S, minerals, e.g. NaCl, industrial waste	Bad smell, not suitable for drinking	

Biological

Impurity/pollutants	Cause	Effect	Remedial measures
1. Pathogenic organisms	Human and animal fecal waste	Water-borne diseases, e.g. cholera, typhoid, paratyphoid, dysentery, diarrhea, vomiting	Disinfection, e.g., by boiling, by ultraviolet rays, by using ozone, potassium per magnate, chlorination

Chemical pollutants

a. Some chemicals

Impurity/pollutants	Cause	Effect	Remedial measures
Acidity/alkalinity (PH) PH = log10[1/H+]	Presence of acid or alkali	Acidic water: tuberculosis, corrosion. Alkaline water: incrustation, sediment deposits	Neutralizing
Calcium and Magnesium	Natural	Hardness	Water softening
Chloride (In the form of NaCl)	Natural pollution from seawater, brine, or industrial and domestic waste	Not significant in small amounts, salty taste, corrosion	Treatment methods, such as dilution, reverse osmosis, distillation
Sulfate	Natural	Not significant in small amount, Laxative effect, hardness, taste	Treatment method, such as reverse osmosis, distillation
Fluoride	Water additive for promoting strong teeth, erosion of natural deposits, fertilizer, and aluminum factories	<1ppm, fewer cavities in the teeth of children >1.5ppm, spotting and discoloration of teeth	<1ppm, fluoridation (adding fluoride compound) >1.5ppm, de-fluoridation (e.g. lime-soda process)

Phosphate	Natural, agriculture, boiler water, laundries	Algal growth	Chemical precipitation

b. Dissolved gas

Impurity/pollutants	Cause	Effect	Remedial measures
Dissolved oxygen	Absorption from the atmosphere	Positive effect: taste. Negative: corrosion	Chemical method for treatment, e.g., using Sodium sulfite, boiling
Dissolved CO_2	Absorption from atmosphere	Bad taste, odor, corrosion	Treatment method, such as aeration
Dissolved H_2S	Natural: due to sulfur-reducing bacteria, hot water heater fitted with Magnesium	Bad taste, odor, corrosion	Treatment methods, such as aeration, activated carbon, oxidation

c. Forms of Nitrogen

Impurity/pollutants	Cause	Effect	Remedial measures
Nitrite, Nitrate	Runoff from fertilizer use, leaching from septic tanks, sewage, and erosion of natural deposits	Effects on infants, blue-baby syndrome, and algal growth	Treatment methods, such as biological treatment, distillation, and reverse osmosis
Ammonia	Metabolic, agriculture	Not of immediate health relevance, Pollution, and growth of algae	Treatment method, such as biological treatment, aeration

d. Agrochemicals

Impurity/pollutants	Cause	Effect	Remedial measures
Pesticide	Agriculture use	Positive: Increase in crop production Negative: water pollution, effects on other animals, birds, including human health	Pest management by eco-friendly means, e.g., crop rotation, multi-crop agriculture, natural predators, parasites, and pathogens for controlling pest, sterilization
Chemical fertilizer	Agriculture use	Positive: Increase in crop production Negative: water pollution, algal growth	Use of organic fertilizers (manure, compost), use of a minimum level of inorganic fertilizer

e. Metals

Impurity/pollutants	Cause	Effect	Remedial measures
Iron and Manganese	Natural deposits, iron pipes	Taste, color, and turbidity, staining of clothes, and incrustation in water mains	Treatment methods, such as aeration, oxidation
Copper	Corrosion of household	Liver or kidney	Corrosion control

	plumbing systems; erosion of natural deposits	damage, effect on lungs, restriction in the growth of aquatic plants	
Zinc	Natural deposits	Not water hazard overdose: vomiting, dizziness	Treatment method, such as coagulation, filtration
Aluminum	Natural deposits, treatment using Aluminum sulfate as coagulant	Neurological disorders	Treatment methods, such as reverse osmosis, softening

f. Some toxic metals

Impurity/ pollutants	Cause	Effect	Remedial measures
Arsenic	Natural, Industrial effluents (glass & electronics), medicinal use	Toxic, respiratory, and skin cancer, nervous disorders	Use of low arsenic water, e.g. rain rainwater Treatment method, such as reverse osmosis, filtration
Lead	Corrosion of plumbing systems, erosion of natural deposits, industrial waste, dust, paint	Kidney problems, high blood pressure, and nervous disorder	Not using water containing lead, Treatment method, such as filtration
Mercury	Erosion of natural deposits, industrial discharge	Highly toxic, Kidney damage, nervous disorder, blurred vision	Treatment method, such as filtration, granular activated Carbon, reverse osmosis
Cadmium Chromium Cyanide Barium	Erosion of natural deposits, industrial discharge	Cadmium: kidney, lung Chromium: respiratory Cyanide: nerve damage Barium: High blood pressure	Treatment method, such as reverse osmosis

3.3 Air pollution

Composition of the atmosphere
N_2: 78%, O_2: 21%, Other gases: 1% e.g. Argon, CO_2, H_2, He, CH_4, O_3, Neon, CO, NO_2, NH_3 etc.

Air pollution: presence of certain substances in the air in high enough concentrations and for long enough duration to cause undesirable effects

Sources of air pollution
1. **Natural sources**
    - Forest fires, dust storms, volcanic eruptions, salt sea spray, pollen grains
2. **Man-made sources**
    - Fuel combustion: coal, gas
    - Automobile emissions
    - Industrial emissions: iron and steel manufacturing, oil refining, brick factory, cement factory, chemical and petrochemical operations, pulp and paper industry, fertilizer plants, thermal power plants, textile industry, etc.
    - Decomposition of organic waste and municipal garbage

Classification of air pollutants based on origin
1. **Primary:** pollutants that are directly emitted to the atmosphere. The primary pollutants
    - SO_2: due to coal burning
    - NO_2: due to combustion of fossil fuels, e.g., coal or gasoline
    - CO: due to incomplete combustion of fossil fuels
    - Particulate matter (solid or liquid droplets, <10 µm, suspended in air)
    - Particulate lead: due to vehicle emissions
2. **Secondary**: pollutants that are formed in the atmosphere by chemical reaction. Main secondary pollutants
    - H_2SO_4: formed by chemical reaction between SO2 and H2O
    - O_3: photochemical reaction between HC and NO

Definitions
Dust: 1 to 100 µm, Smoke or fume: less than 1 µm, Mist: suspension of liquid particles between 0.1 to 10 µm, Spray: liquid particles >10 µm, Smog: Smoke and fog

Hazardous or toxic air pollutants
- **Asbestos:** due to the demolition of old buildings containing Asbestos fireproofing, cancer, and lung disease
- **Benzene:** due to gasoline-powered vehicles, cancer
- **Beryllium**: from foundries, ceramic factories, incinerations
- **Mercury:** coal burning, incineration of garbage
- Vinyl Chloride
- Radioactive air pollutants

Effects of air pollution
1. **Health effects**
    - Chronic disease, Respiratory illness: bronchitis, asthma, lung cancer
    - Temporary effect: nose or eye or throat irritation, coughing, chest pain, general discomfort

2. **Damage to material objects**
    - Soiling and deterioration of building surface, corrosion of metals, weakening of rubber, textile, and synthetic materials
3. **Effect on vegetation**
    - damage to trees, flowers, fruits, and vegetables
4. **Effects on the physical properties of the atmosphere**
    - Effects on visibility
    - Effects on urban atmosphere and weather conditions: fog, cloud, precipitation
    - Effects on atmospheric constituents: increase in atmospheric CO2

Air pollution control

1. Natural self-cleansing of the environment: dispersion by wind, settling by gravity, washout by rain, adsorption by soil, rocks, leaves, and buildings
2. Control of particulate pollutants in industries using a mechanical device
3. Control of gaseous pollutants in industries using a mechanical device
4. Controlling air pollution from automobiles
5. Air quality legislation and standards

Controlling air pollution from automobiles

(CO, HC, NO_x, particulates, SO_2)

- **Catalytic converter**: for complete oxidation of combustible fuel
- Reducing lead and sulfur content in gasoline
- Correct operation and maintenance of the engine
- **Fuel substitutions**: use of reformulated gasoline (oxygenated fuel containing at least 2% of O_2) or alternate fuels such as liquefied petroleum gas (LPG), compressed natural gas (CNG), methanol, ethanol, propane, Hydrogen, electric-powered vehicle

Indoor air pollution

- Pollution of the air inside buildings

Air exchange methods: **infiltration** (through cracks, joints, holes), **natural ventilation**, **forced ventilation** (e.g., fans)

Sources of indoor air pollution

- Combustion products: tobacco smoke, combustion from stove, heater, fireplace, chimney
- Asbestos: fire-resistance and insulation in buildings
- Radon: Radioactive decay of Radium found in soil and rock
- Organic chemicals from household products: paints, waxes, varnishes, cleaning agents, pesticides, cosmetics, hobby materials
- Formaldehyde: used in paints, coatings, glues, adhesives
- Lead: paint, dust
- Biological substances: bacteria, fungi, viruses, house dust,

pollen.

Effects of indoor air pollution

- Health problems: eye, nose, and throat irritation, respiratory problems, headache, dizziness, visual problems, memory impairment, asthma, cancer, transmission of infectious disease, e.g., influenza, measles, etc.

Remedial measures of indoor air pollution

- Proper ventilation, use of fans, exhaust fans, inspection of chimneys, furnaces annually, restricted use of asbestos, and proper disposal of organic materials

3.4 Noise pollution

Sound: produced by mechanical vibration of a sound source, transmitted in the form of a wave.
Wavelength: distance between peaks or valleys
Amplitude: height of the peak of the wave
Frequency: no. of wavelengths in 1S (cycle/s or HZ)
Single wavelength: cycle
Decibel scale for sound

Noise pollution: unwanted sound that produces undesirable physiological and psychological effects.

Source

- Traffic: air traffic, road traffic, seashore, and inland water traffic
- Industries
- Others: loudspeaker, siren, shouting, ringing bell, general daily activities

Effect

- General discomfort
- Reduction in the efficiency of people
- Psychological effect
- Effect on sleep, recreation, and personal communication
- Reduction in gastric activity, dizziness, and a rise in breathing
- Irritation, anxiety, and stress
- Lack of concentration
- Mental fatigue
- Effect of prolonged exposure: Physical damage to the ear, temporary/permanent hearing loss, or nervous breakdown, increase in blood pressure

Countermeasures

- Protection of the recipient: use of air plugs or air muffs
- Increasing distance
- Noise barriers: absorptive materials, e.g., heavy drapes, carpets, special ceiling, wall acoustic material
- Reduction of noise at the source
- Rules and regulations

3.5 Global warming (Greenhouse effect)

Global warming

- Rise in global mean temperature of the Earth

Solar energy: short-wave radiation
Energy radiated from the Earth's surface: long-wave radiation

Greenhouse effect

- Concept of conventional greenhouses with glass transmits short-wave radiation, opaque to long-wave radiation
- Greenhouse effect: effect caused by greenhouse gases in the atmosphere in which short-wave radiation is transmitted to the earth's surface, but the long-wave radiation from the earth is absorbed, thereby increasing the temperature.

Greenhouse gases

- A group of about 20 gases is responsible for the greenhouse effect through their ability to absorb long-wave terrestrial radiation
- occupy less than 1% of the total volume of the atmosphere
- **Major greenhouse gases**
  - CO_2: major, responsible for 60% of total GHG
  - CH_4
  - NOx, mainly N_2O: responsible for 7% of total GHG
  - Chlorofluorocarbons (CFC): responsible for 25% of total GHG
  - O_3
  - Water vapor

Cause of global warming: Enhancement of green greenhouse effect due to anthropogenic activities. Sources of GHG

- CO_2: Burning of fossil fuels (oil, gas, and coal), large-scale deforestation
- CH_4: large-scale decomposition of organic matter in swamps, rice paddy, livestock yards, cattle raising, biomass burning
- N_2O: soil and fertilizer, groundwater and oceans, combustion
- CFC: using refrigerant, air-conditioning, fire extinguisher, cleaning solvent, blowing agent, aerosol spray
- O_3: upper natural environment

Prediction of global warming

- Using global circulation models (GCM): computer analysis of mathematical equations that model Earth's atmosphere

Impact of global warming

- Rise in temperature: 0.3 to 0.6 °C in the last century
- Sea level rise: due to thermal expansion of water on oceans and melting ice caps and glaciers, 1-2 mm/year over the last century, flooding of coastal areas, beach erosion, saltwater intrusion into coastal areas
- Effect on water resources: change in the pattern of evaporation and precipitation, increase in evaporation and precipitation, more precipitation in the form of rain, increase in runoff

- Effect on storms and desertification: more storms, expansion of deserts and sub-arid areas with higher evaporation
- Socio-economic effect: chances of disease due to high temperature, increase in poverty due to flood and drought
- Ecological effect: effect on agriculture and forest ecosystems

Countermeasures

- Environmental taxes on GHG emissions
  Using the revenue from taxes to develop permanent and stable funding for improved efficiency, and developing renewable energy sources

International efforts to control global warming

1. **Atmospheric scientists meeting in Geneva in 1990**
   - Steps to reduce emission of GHGs: industrialized nations could reduce CO2 emissions by 20% by 2005
2. **Earth Summit in Rio de Janeiro in 1992**
   - Signing of the treaty to stabilize emissions of GHGs at the year 1990 level by the year 2000
3. **Global warming conference in Berlin in 1995**
   - Binding timetable for reduction in GHG emissions after the year 2000
4. **International conference in Kyoto, Japan, in 1997**
   - Kyoto protocol: a set of binding emission targets and timelines for developed nations

3.6 Acid rain

PH: measure of H ion concentration, range: 0-14, 7: neutral, <7: acidic, <7: alkaline.
Rainwater: naturally acidic, with a pH of about 5.6
Acid rain: Rainwater with a pH <5.6 that results from air pollution caused by human activities

Causes

- **Emission of SO_2 and NO_x into the atmosphere**
  - Natural source: decomposition and forest fire, volcanic eruptions
  - Anthropogenic: burning of fossil fuels, industrial processes, and gasoline-powered automobiles
- **Transformation into mild sulfuric or nitric acid by combining with water vapor**
- Dissolution of H_2SO4, HNO_3, and oxides of Nitrogen and Sulfur, and other gases in clouds containing rain and settling down of acid rain

Wet deposition: the pollutant material that comes down with rain, including particulates and gases
Dry deposition: the material reaching the ground by gravity during dry intervals, includes particulates and gases, and aerosols

Impact of acid rain

- Lowering of PH in lakes and rivers, springs, and wells, harming fish and aquatic life
- Decline in forest, reduction in pollination of crops, crop quality, and quantity
- Deterioration of building materials, e.g., steel, paint, plastics, cement, masonry, limestone, marble, sandstone
- Potential infiltration to groundwater and an increase in the solubility of toxic materials (Pb, Cu, Zn) in groundwater

- Effect on human health: due to acidic surface and groundwater consumption, respiratory illness, and asthma
- Corrosion of water pipes, dissolving metals, e.g., lead, copper, and iron in water pipes, causing direct harm to humans through consumption
- Damage to soil fertility

Countermeasures

a. Technological approaches

1. pre-combustion: choose fuel with low S and N content or treat the fuels, physical and chemical process to remove S and N
2. Reduce emission of pollutants during combustion, e.g., catalytic or coal-limestone combustion
3. post-combustion: reduce emissions by high efficiency removal techniques, e.g., scrubber

b. Environmental clean-up and restoration

e.g., liming of acidified surface water bodies (for neutralization) to save or restore many important resources

c. Technical measure to reduce CO_2 emissions

1. Improve the efficiency of fuel to usable ends
2. Direct removal of CO_2: technique for removal of CO_2 from atmosphere by power plant
3. Reduction of CO_2 by forestry
4. Cleaner energy production, e.g., photovoltaic, wood, or wind

3.7 Ozone depletion

- Important role concerning atmospheric chemistry in both the troposphere and the stratosphere
- Pollutants at ground level, but stratospheric O3 is crucial for life on the earth: blocks/absorbs most of the harmful ultraviolet (UV) rays coming from the sun, thus protecting plants and animals

Effect of UV

- Human skin cancer, eye cataracts, suppression of the immune system response
- Effect on plants and aquatic life

Ozone hole: ozone-depleted region over Antarctica

O_3: unstable molecule, balance between formation and removal

Main cause of O_3 depletion: presence of Chlorofluorocarbons (CFCs) in the atmosphere

Source of CFC: using refrigerant, air-conditioning, fire extinguisher, cleaning solvent, blowing agent, aerosol spray

From CFC, the release of Cl atom by UV acts as a catalyst for the destruction of O_3

Countermeasures

- Adoption of environmentally safe alternatives to CFCs for refrigeration and thermal insulation
- Reduction in CFC use